

Fundamentals of theory of computation I.

Lecture 3

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Linear Grammars

Definition

A context-free grammar $G = (N, \Sigma, P, S)$ is called **linear** if all rules are either of the form

- a) $A \rightarrow u, A \in N, u \in \Sigma^*$ or
- b) $A \rightarrow u_1 B u_2$, where $A, B \in N$ and $u_1, u_2 \in \Sigma^*$.

A linear grammar G is called **left linear** if we have $u_1 = \varepsilon$ for all of its rules of type b).

A linear grammar G is called **right linear** if we have $u_2 = \varepsilon$ for all of its rules of type b).

Observation: Right linear grammar is just another name for regular (type 3) grammar.

Example:

- 1. $S \rightarrow aSb \mid \varepsilon$ linear, but neither left or right linear
- 2. $S \rightarrow Saa \mid b$ left linear
- 3. $S \rightarrow aS \mid bbS \mid c$ right linear

Linear grammars and languages

Definition

A language L is called linear/left linear/right linear if there is a linear/left linear/right linear grammar G with the property $L = L(G)$.

Remark: So, a right linear language is another term for regular (type 3) languages.

Theorem

Every left linear grammar generates a regular language.

Left and right linear languages

Proof: Let $G = (N, \Sigma, P, S)$ be a left linear grammar and let $N = \{S, A_1, \dots, A_n\}$.

Assume, that S does not occur on the right hand side of any rule. (In the case it occurs, rename all occurrences of S to S' , $S' \notin N \cup \Sigma$ and add the rule $S \rightarrow S'$ to P .)

We construct a right linear grammar $G' = (N, \Sigma, P', S)$ with the property $L(G) = L(G')$ as follows.

1. $S \rightarrow u : \in P'$ if and only if $S \rightarrow u \in P$,
2. $S \rightarrow uA_k : \in P'$ if and only if $A_k \rightarrow u \in P$,
3. $A_j \rightarrow uA_k : \in P'$ if and only if $A_k \rightarrow A_ju \in P$,
4. $A_j \rightarrow u : \in P'$ if and only if $S \rightarrow A_ju \in P$.

$(u \in \Sigma^*)$

We have to prove, that $L(G) = L(G')$.

Left and right linear languages

$L(G) \subseteq L(G')$:

Let $w \in L(G)$.

$S \rightarrow u : \in P'$	$\leftrightarrow$	$S \rightarrow u \in P,$
$S \rightarrow uA_k : \in P'$	$\leftrightarrow$	$A_k \rightarrow u \in P,$
$A_j \rightarrow uA_k : \in P'$	$\leftrightarrow$	$A_k \rightarrow A_j u \in P,$
$A_j \rightarrow u : \in P'$	$\leftrightarrow$	$S \rightarrow A_j u \in P.$

Case 1 If $S \rightarrow w \in P$, then $S \rightarrow w \in P'$, so $w \in L(G')$ holds.

Case 2 Otherwise, there exists a derivation

$$S \Rightarrow A_{i_1} w_1 \Rightarrow A_{i_2} w_2 w_1 \Rightarrow \cdots \Rightarrow A_{i_{m-1}} w_{m-1} \cdots w_1 \Rightarrow w_m w_{m-1} \cdots w_1 = w$$

in G of w ($m \geq 2$). Then we have a derivation

$$S \Rightarrow w_m A_{i_{m-1}} \Rightarrow w_m w_{m-1} A_{i_{m-2}} \Rightarrow \cdots \Rightarrow w_m \cdots w_2 A_{i_1} \Rightarrow w_m \cdots w_1 = w$$

in G' proving $w \in L(G')$.

$L(G') \subseteq L(G)$:

It follows from symmetry.

Left and right linear languages

Example:

$$L = \{a^n b^k \mid n, k \in \mathbb{N}\}.$$

$S \rightarrow u : \in P'$	$\leftrightarrow$	$S \rightarrow u \in P,$
$S \rightarrow uA_k : \in P'$	$\leftrightarrow$	$A_k \rightarrow u \in P,$
$A_j \rightarrow uA_k : \in P'$	$\leftrightarrow$	$A_k \rightarrow A_j u \in P,$
$A_j \rightarrow u : \in P'$	$\leftrightarrow$	$S \rightarrow A_j u \in P.$

$$\begin{array}{l}
 P \\
 S \rightarrow \varepsilon \\
 S \rightarrow A_1 b \\
 S \rightarrow A_2 a \\
 A_1 \rightarrow A_1 b \\
 A_1 \rightarrow A_2 \\
 A_2 \rightarrow A_2 a \\
 A_2 \rightarrow \varepsilon
 \end{array}$$

$$\begin{array}{l}
 P' \\
 S \rightarrow \varepsilon \\
 A_1 \rightarrow b \\
 A_2 \rightarrow a \\
 A_1 \rightarrow bA_1 \\
 A_2 \rightarrow A_1 \\
 A_2 \rightarrow aA_2 \\
 S \rightarrow A_2
 \end{array}$$

$$S \Rightarrow A_1 b \Rightarrow A_2 b \Rightarrow A_2 ab \Rightarrow ab.$$

$$S \Rightarrow A_2 \Rightarrow aA_2 \Rightarrow aA_1 \Rightarrow ab.$$

$\mathcal{L}_3$ is closed under reversal

Proposition

For every right linear grammar $G = (N, \Sigma, P, S)$ there exists a left linear grammar G' generating the reversal of $L(G)$.

Proof: Replace the rules of form $A \rightarrow u$ and $A \rightarrow uB$ by the rules $A \rightarrow u^R$ and $A \rightarrow Bu^R$, respectively ($A, B \in N, u \in \Sigma^*$).

Corollary

1. $\mathcal{L}_3$ is closed under the reversal operation.
2. All regular languages can be generated by a left linear grammar.

Proof:

1. According to the proposition, if $L \in \mathcal{L}_3$, then L^R can be generated by a left linear grammar G . By the previous theorem we have $L^R \in \mathcal{L}_3$.
2. If $L \in \mathcal{L}_3$, then by 1. we have $L^R \in \mathcal{L}_3$, so L^R can be generated by a right linear grammar. According to the proposition $(L^R)^R = L$ can be generated by a left linear grammar.

Type 3 normal form

Definition

A grammar $G = (N, \Sigma, P, S)$ is of **type 3 normal form**, if all rules of P is either of the form

- ▶ $X \rightarrow aY$, $X, Y \in N, a \in \Sigma$ or
- ▶ $X \rightarrow \varepsilon$, where $X \in N$.

Theorem

All regular languages can be generated by a type 3 normal form grammar.

Proof: Let L be a regular language. By definition, there exists a grammar $G = (N, \Sigma, P, S)$ of type 3, such that $L(G) = L$ holds. The rules of G are either of the form

- a) $A \rightarrow uB$ or
- b) $A \rightarrow u$ ($A, B \in N, u \in \Sigma^*$).

We transform G to the appropriate form in 2 steps taking care of transformation steps should not change the generated language.

Type 3 normal form

Step 1 (Length reduction)

for rules of type a):

If $A \rightarrow uB \in P$ ($A, B \in N, u \in \Sigma^*$), $|u| > 1$ and $u = a_1 \cdots a_n, n \geq 2$, then replace the rule $A \rightarrow a_1 \cdots a_n B$ by the set of rules $\{A \rightarrow a_1 Z_1, Z_1 \rightarrow a_2 Z_2, \dots, Z_{n-1} \rightarrow a_n B\}$, where $Z_1, \dots, Z_{n-1}$ are new nonterminals, they can not be reused for the reduction of other long rules.

for rules of type b):

Similarly, if $A \rightarrow u \in P$ ($A \in N, u \in \Sigma^*$), $|u| > 0$ and $u = a_1 \cdots a_m, m \geq 1$, then replace the rule $A \rightarrow a_1 \cdots a_m$ by the set of rules $\{A \rightarrow a_1 Y_1, Y_1 \rightarrow a_2 Y_2, \dots, Y_{m-1} \rightarrow a_m E, E \rightarrow \varepsilon\}$ where $Y_1, \dots, Y_{m-1}$ are new nonterminals, they can not be reused for the reduction of other long rules.

E is a new nonterminal as well, but it can be used commonly for all the length reductions of type b) rules.

Type 3 normal form

Consider a derivation in G applying a long rule. Replace it by the corresponding sequence of short rule applications simulating the application of the long rule. Due to the uniqueness of the new nonterminals no other terminal word can be derived after the transformation.

So, length reduction does not change the generated language.

Step 2 (Chain elimination)

Let $G = (N, \Sigma, P, S)$ be the grammar after length reduction.

So, rules of P are of the form $X \rightarrow aY$, $X \rightarrow Y$, $X \rightarrow \varepsilon$ where $X, Y \in N$ and $a \in \Sigma$.

The only remaining task is to eliminate **chain rules**, i.e. rules of the form $X \rightarrow Y$, where $X, Y \in N$.

Type 3 normal form

Let P_0 be the set of chain rules of P .

For all $A \in N$ we determine the set $H(A) := \{B \in N \mid A \Rightarrow^* B\}$.

To determine $H(A)$ efficiently, define the sets $H_i(A)$ ($i \geq 0$) recursively as follows.

$$H_0(A) := \{A\}$$

$$H_{i+1}(A) := H_i(A) \cup \{B \in N \mid \exists C \in H_i(A) \text{ such that } C \rightarrow B \in P\}.$$

$$H_0(A) \subseteq H_1(A) \subseteq \cdots \subseteq H_k(A) \subseteq \cdots \subseteq N$$

$$k := \min\{0 \leq i \leq n - 1 \mid H_i(A) = H_{i+1}(A)\}.$$

Then $H(A) = H_k(A)$, since $H_i(A)$ contains exactly those nonterminals, that are derivable from A in at most i steps ($i \geq 0$).

Type 3 normal form

$$P' := \{A \rightarrow w \mid \exists B \in H(A), \text{ such that } B \rightarrow w \in P\} \setminus P_0.$$

Consider a derivation, and assume that it contains some consecutive chain rule applications from A to B followed by an application of a rule $B \rightarrow w$. Then, in the corresponding derivation in $G' = (N, \Sigma, P', S)$ we can replace $A \Rightarrow^* B \Rightarrow w$ by $A \Rightarrow w$.

Doing this transformation in the opposite way one can see, that for every derivation in G' there exist a corresponding derivation in G generating the same word.

So G' generates the same language and it is of type 3 normal form.

Type 3 normal form

Example:

$$S \rightarrow abS$$

$$S \rightarrow B$$

$$B \rightarrow bB$$

$$B \rightarrow V$$

$$V \rightarrow aa$$

$$V \rightarrow b$$

Step 1 Length reduction

$$S \rightarrow aZ_1 \quad Z_1 \rightarrow bS$$

$$S \rightarrow B$$

$$B \rightarrow bB$$

$$B \rightarrow V$$

$$V \rightarrow aZ_2 \quad Z_2 \rightarrow aE$$

$$V \rightarrow bE \quad E \rightarrow \varepsilon$$

Step 2 Chain elimination

$$S \rightarrow aZ_1 \quad Z_1 \rightarrow bS$$

$$S \rightarrow B$$

$$B \rightarrow bB$$

$$B \rightarrow V$$

$$V \rightarrow aZ_2 \quad Z_2 \rightarrow aE$$

$$V \rightarrow bE \quad E \rightarrow \varepsilon$$

$$S \rightarrow aZ_1 \quad Z_1 \rightarrow bS$$

$$S \rightarrow bB$$

$$S \rightarrow aZ_2 \quad S \rightarrow bE$$

$$B \rightarrow bB$$

$$B \rightarrow aZ_2 \quad B \rightarrow bE$$

$$V \rightarrow aZ_2 \quad Z_2 \rightarrow aE$$

$$V \rightarrow bE \quad E \rightarrow \varepsilon$$

$$H_0(X) = H_1(X) = H(X) = \{X\} \quad (X \in \{V, Z_1, Z_2, E\})$$

$$H_0(B) = \{B\}, H_1(B) = H_2(B) = H(B) = \{B, V\}.$$

$$H_0(S) = \{S\}, H_1(S) = \{S, B\}, H_2(S) = H_3(S) = H(S) = \{S, B, V\}.$$

Regular expressions

Motivation:

Observe, that all finite languages are regular. It was proved earlier, that $\mathcal{L}_3$ (the class of regular languages) is closed under regular operations (union, concatenation, iterative closure).

So, starting from some elementary, finite languages and applying finitely many regular operations (union, concatenation, iterative closure) on them no non-regular language can be produced.

Is it true, that all regular languages can be produced this way?

These finite constructions consisting of elementary languages and regular operations are called regular expressions.

So, are regular expressions strong enough to describe the language class $\mathcal{L}_3$?

Regular expressions

Definition

Let V and $V' = \{\emptyset, \varepsilon, \cdot, +, *, (,)\}$ be disjoint alphabets. The set of **regular expressions** over V is the minimal set defined by the following recursion.

1. $\emptyset$ is a regular expression over V ,
2. ε is a regular expression over V ,
3. a is a regular expression over V for every $a \in V$,
4. if R is a regular expression over V then R^* is a regular expression over V as well,
5. if Q and R are regular expressions over V , then $(Q \cdot R)$ is a regular expression over V as well,
6. if Q and R are regular expressions over V , then $(Q + R)$ is a regular expression as well.

Regular expressions

Definition

Every regular expression R over V **determines** (or **denotes**) a language $L(R) \subseteq V^*$ defined recursively as follows.

1. $L(\emptyset) := \emptyset$,
2. $L(\varepsilon) := \{\varepsilon\}$,
3. $L(a) := \{a\} \quad (a \in V)$,
4. $L(R^*) := L(R)^*$, where R is a regular expression over V ,
5. $L(Q \cdot R) := L(Q)L(R)$, where Q and R are a regular expressions over V ,
6. $L(Q + R) := L(Q) \cup L(R)$, where Q and R are a regular expressions over V .

So, regexp operations $*$, $\cdot$, $+$ correspond to $*$, concatenation, union, respectively.

Alternative notations: (QR) instead of $(Q \cdot R)$.

$(Q | R)$ or $(Q \cup R)$ instead of $(Q + R)$.

Regular expressions

Remark: Shortly, but somewhat imprecisely, R can be used instead of $L(R)$ in several situations.

E.g., $ab^* = \{ab^n \mid n \in \mathbb{N}\}$ instead of $L(ab^*) = \{ab^n \mid n \in \mathbb{N}\}$.

Claim: Let P, Q, R be regular expressions. Then we have

1. $P + (Q + R) = (P + Q) + R$
2. $P \cdot (Q \cdot R) = (P \cdot Q) \cdot R$
3. $P + Q = Q + P$
4. $P \cdot (Q + R) = P \cdot Q + P \cdot R$
5. $(P + Q) \cdot R = P \cdot R + Q \cdot R$
6. $P^* = \varepsilon + P \cdot P^*$
7. $\varepsilon \cdot P = P \cdot \varepsilon = P$
8. $P^* = (\varepsilon + P)^*$
9. $P \cdot \emptyset = \emptyset \cdot P = \emptyset$

(More precisely, $L(\text{left hand side}) = L(\text{right hand side})$ holds in these cases.)

Regular expressions

Order of (decreasing) precedence: $*$, $\cdot$, $+$.

Due to associativity of union and concatenation and our agreement on the order of precedence, some pairs of parentheses might be omitted.

Example:

Regular expressions $(a + b)a^*$ and $aa^* + ba^*$ denote the same language $\{aa^n \mid n \in \mathbb{N}\} \cup \{ba^n \mid n \in \mathbb{N}\}$.

On the other hand $(a + b)a^* \neq a + ba^*$, since the latter describes $\{a, b, ba, ba^2, ba^3, \dots\}$.

Remark: The following abbreviations can be used to get a shorter expression. Q^+ stands for $(Q \cdot Q^*)$. **Specific** positive integer powers, like in $(a^3b + ab^3)b^*$.

Regular expressions – Arden's theorem

The following theorem will be used later, at studying the connection between regular expressions and finite automata.

Arden's theorem

Given two regular expressions R and Q . Consider the problem of determining a regular expression P satisfying the equation $P = R + P \cdot Q$. It has a solution $P = R \cdot Q^*$ and this solution is unique if $\varepsilon \notin Q$ holds.

Example: $P = ab^* + Pbc^*$ has a unique solution $P = ab^*(bc^*)^*$.

Proof:

(1) $P = R \cdot Q^*$ is a solution:

$$\begin{aligned} R + P \cdot Q &= R + (R \cdot Q^*) \cdot Q = R + R \cdot (Q \cdot Q^*) = \\ &= R \cdot (\varepsilon + Q \cdot Q^*) = R \cdot Q^* = P. \end{aligned}$$

Regular expressions – Arden's theorem

(2) Uniqueness of the solution:

$P = R + P \cdot Q = R + (R + P \cdot Q) \cdot Q = R + RQ + PQ^2 = \dots = R + RQ + \dots RQ^n + PQ^{n+1}$, where n is an arbitrary natural number, so we have

$$P = R(\varepsilon + Q + \dots + Q^n) + PQ^{n+1}. \quad (*)$$

Let $w \in P$ and consider (*) for $n = |w|$. Since $\varepsilon \notin Q$ we have $w \notin PQ^{n+1}$ (each word of PQ^{n+1} is of length at least $n + 1$), implying $w \in R(\varepsilon + Q + \dots + Q^n) \subseteq RQ^*$.

For the other direction, if $w \in RQ^*$, then $\exists n \in \mathbb{N}$, such that $w \in RQ^n$, i.e., w is in the right hand side of (*). This implies $w \in P$.

So, we have proved uniqueness of the solution $P = R \cdot Q^*$ in the case $\varepsilon \notin Q$.

Descriptive power of regular expressions

Theorem

Every regular expression determines a regular (i.e., type 3) language and for every regular language there exists a regular expression determining that language.

Proof:

1. Regular expressions are denoting regular languages since $\emptyset, \{\varepsilon\}, \{a\}$ ($a \in V$) are regular and $\mathcal{L}_3$ is closed under regular operations.
2. We prove, that for every regular language L generated by a grammar $G = (N, \Sigma, P, A_1)$ in type 3 normal form there exists a regular expression R such that $L(R) = L$.

Descriptive power of regular expressions

Let $N = \{A_1, \dots, A_n\}$, $n \geq 1$. A_1 is the initial symbol of G and all rules are of the form $A_i \rightarrow aA_j$ or $A_i \rightarrow \varepsilon$, where $a \in \Sigma$, $1 \leq i, j \leq n$.

We say, that a derivation $A_i \Rightarrow^* uA_j$ ($u \in \Sigma^*$) **involves** a nonterminal A_m if A_m occurs in an intermediate word in this derivation of uA_j from A_i .

A derivation $A_i \Rightarrow^* uA_j$ is called **ℓ -restricted** if $1 \leq m \leq \ell$ holds for the nonterminals A_m involved in the derivation.

Define the sets

$$E_{i,j}^k = \{u \in \Sigma^* \mid \text{there exists a } k\text{-restricted derivation } A_i \Rightarrow^* uA_j\}$$

for all $0 \leq k \leq n$, $1 \leq i, j \leq n$.

Descriptive power of regular expressions

For $i \neq j$ the set $E_{i,j}^0$ is either $\emptyset$ or a set of letters of Σ .

$$a \in E_{i,j}^0 \text{ if and only if } A_i \rightarrow aA_j \in P.$$

If $i = j$, then $E_{i,i}^0$ contains ε in addition since $A_i \Rightarrow^* A_i$ holds.

The proof is by induction on k .

For all $1 \leq i, j \leq n$ $E_{i,j}^0$ can be denoted by a regular expression.

Assume, for a given $1 \leq k \leq n$, that $E_{i,j}^{k-1}$ can be denoted by a regular expression for all $1 \leq i, j \leq n$. We have

$$E_{i,j}^k = E_{i,j}^{k-1} + E_{i,k}^{k-1} \cdot (E_{k,k}^{k-1})^* \cdot E_{k,j}^{k-1}.$$

(Consider the occurrences of A_k 's in a k -restricted derivation of a word $u \in E_{i,j}^k$. The part before the first occurrence, the intermediate parts and the part after the last occurrence are all $(k-1)$ -restricted.)

Descriptive power of regular expressions

Since $E_{i,j}^k$ can be expressed by $(k - 1)$ -restricted sets and regular operations we have, by induction, that $E_{i,j}^k$ can be denoted by a regular expression for all $1 \leq i, j \leq n, 0 \leq k \leq n$.

Let $I_\varepsilon = \{1 \leq i \leq n \mid A_i \rightarrow \varepsilon \in P\}$. Then $L(G) = \bigcup_{i \in I_\varepsilon} E_{1,i}^n$ proving that L can be denoted by a regular expression.

Descriptive power of regular expressions

Example 1:

$R = a(a + ba^*)^*$. Make a type 3 grammar generating $L(R)$.

We use the constructions in the proof of our earlier theorem on $\mathcal{L}_3$ being closed under regular operations.

Let $R_1 = a$, $R_2 = ba^*$, $R_3 = R_1 + R_2$, $R_4 = R_3^*$.

$R_1 : A \rightarrow a$

$R_2 : B \rightarrow bC, C \rightarrow aC \mid \varepsilon$ (start symbol: B)

$R_3 : D \rightarrow A \mid B, A \rightarrow a, B \rightarrow bC, C \rightarrow aC \mid \varepsilon$
(start symbol: D)

$R_4 : X \rightarrow D \mid \varepsilon, D \rightarrow A \mid B, A \rightarrow a \mid aD, B \rightarrow bC, C \rightarrow aC \mid \varepsilon \mid D$
(start symbol: X)

$R : S \rightarrow aX, X \rightarrow D \mid \varepsilon, D \rightarrow A \mid B, A \rightarrow a \mid aD, B \rightarrow bC,$
 $C \rightarrow aC \mid \varepsilon \mid D$ (start symbol: S)

Descriptive power of regular expressions

Example 2:

$$A_1 \rightarrow bA_1 \mid aA_2$$

$$A_2 \rightarrow bA_1 \mid \varepsilon$$

$$E_{i,j}^k = E_{i,j}^{k-1} + E_{i,k}^{k-1} \cdot (E_{k,k}^{k-1})^* \cdot E_{k,j}^{k-1}$$

$E_{i,j}^0$	$j = 1$	$j = 2$
$i = 1$	$\varepsilon + b$	a
$i = 2$	b	ε

$E_{i,j}^1$	$j = 1$	$j = 2$
$i = 1$	b^*	b^*a
$i = 2$	b^+	$\varepsilon + b^+a$

$E_{i,j}^2$	$j = 1$	$j = 2$
$i = 1$	$b^* + b^*a(b^+a)^*b^+$	$b^*a(b^+a)^*$
$i = 2$	$(b^+a)^*b^+$	$(b^+a)^*$

E.g.,

$$E_{12}^2 = E_{12}^1 + E_{12}^1(E_{22}^1)^*E_{22}^1 =$$

$$b^*a + b^*a(\varepsilon + b^+a)^*(\varepsilon + b^+a) = b^*a + b^*a(b^+a)^* = b^*a(b^+a)^*.$$

$$L(G) = E_{12}^2 = b^*a(b^+a)^*.$$